

TOKAT, Turkey

WMO: 170860

Lat: 40.3311N

Lon: 36.5578E

Elev: 1995

StdP: 13.67

Time Zone: 3.00 (E03)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.3	20.7	8.5	9.1	20.2	12.4	11.1	25.7	18.9	52.1	15.8	46.0	4.1	90	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.2	94.4	67.6	90.6	66.8	87.5	66.1	70.7	89.1	69.1	86.0	67.7	83.3	6.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	98.6	78.5	63.1	93.2	77.1	61.8	88.9	76.1	35.9	88.7	34.5	86.5	33.2	83.5	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.2	13.2	11.6	10.0	101.7	7.3	2.4	4.7	103.4	0.5	104.8	-3.6	106.2	-9.0	108.0		
			9.1	72.6	7.1	2.8	4.0	74.6	-0.1	76.2	-4.1	77.8	-9.3	79.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.3	37.0	40.4	47.3	55.3	63.0	69.4	73.7	75.2	68.2	59.1	46.9	38.5
	DBStd	15.11	8.73	9.16	8.21	7.42	6.42	4.90	5.06	4.90	6.05	6.67	7.07	8.38
	HDD50	1391	407	284	151	32	2	0	0	0	0	10	141	364
	HDD65	4141	869	689	549	299	115	14	1	1	37	201	544	822
	CDD50	3673	3	15	68	191	405	583	734	781	547	292	47	7
	CDD65	948	0	0	1	8	53	147	270	317	134	18	0	0
	CDH74	9227	0	0	20	154	629	1399	2608	2979	1250	187	1	0
	CDH80	3548	0	0	3	27	181	450	1114	1310	433	30	0	0
Wind	WSAvg	4.8	4.4	5.1	5.5	5.3	4.9	5.0	5.5	5.6	4.9	4.3	3.6	3.9
Precipitation	PrecAvg	16.0	1.8	1.3	1.4	2.1	2.2	1.5	0.4	0.2	0.7	1.4	1.7	2.0
	PrecMax	20.4	4.6	3.3	3.4	4.1	4.3	3.7	2.7	1.4	2.0	5.6	4.5	4.1
	PrecMin	12.3	0.2	0.2	0.1	0.5	0.7	0.1	0.0	0.0	0.0	0.0	0.0	0.4
	PrecStd	2.4	1.1	0.7	0.8	1.0	0.9	0.8	0.6	0.3	0.4	1.2	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.7	66.4	76.4	83.4	90.5	93.0	99.4	99.0	94.6	83.6	71.2	62.0
		MCWB	46.4	49.5	54.2	57.4	64.4	66.1	68.3	69.6	63.9	61.1	53.6	48.3
	2%	DB	55.2	62.2	70.0	78.5	85.1	88.5	94.3	94.6	88.9	78.9	66.6	57.6
		MCWB	44.2	47.9	51.4	55.7	62.2	65.3	68.2	68.5	63.3	59.9	52.6	46.5
	5%	DB	52.3	57.7	65.5	74.1	81.1	85.1	89.9	90.6	84.8	75.2	62.6	53.8
		MCWB	42.6	45.8	49.3	54.0	60.5	64.8	67.4	68.0	62.7	58.7	50.3	44.6
	10%	DB	49.4	54.3	61.7	70.1	77.0	81.9	86.1	87.3	81.3	71.4	58.8	50.5
		MCWB	41.4	43.9	48.0	52.9	59.5	63.6	65.8	67.2	61.9	57.4	48.7	43.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.4	51.4	55.6	61.1	67.3	70.0	73.3	72.9	69.4	64.3	56.0	49.6
		MCDB	57.4	62.7	72.9	78.4	86.4	86.7	92.3	93.5	87.7	78.5	67.2	58.0
	2%	WB	45.0	48.5	52.2	57.6	64.3	67.8	70.7	70.9	66.4	61.4	53.6	47.7
		MCDB	52.9	59.2	67.0	73.1	81.4	83.5	88.9	90.1	81.3	74.0	64.1	55.1
	5%	WB	43.5	46.4	50.2	55.5	62.0	66.3	68.9	69.4	64.7	59.9	51.3	45.7
		MCDB	50.5	56.7	62.9	70.8	77.4	80.8	85.5	86.7	79.5	71.9	59.8	52.6
	10%	WB	42.1	44.5	48.4	53.7	60.1	64.7	67.2	68.0	63.0	58.3	49.4	43.9
		MCDB	48.5	53.4	59.9	67.1	73.9	78.6	82.3	84.0	77.2	69.2	57.7	49.9
Mean Daily Temperature Range	5% DB	MDBR	12.9	15.3	17.9	21.8	23.0	22.9	23.0	23.2	23.7	20.0	17.9	12.6
		MCDBR	16.5	20.8	24.8	29.5	29.9	27.8	29.5	28.7	29.6	26.4	23.4	16.6
		MCWBR	9.1	10.4	11.8	12.7	12.3	10.2	10.1	9.7	10.4	11.0	12.0	9.2
	5% WB	MCDBR	14.5	19.3	22.4	25.7	27.2	24.6	25.9	25.7	25.1	22.4	20.7	15.3
MCWBR		8.4	9.9	11.3	12.3	12.4	10.0	10.6	10.1	10.0	10.1	11.1	8.8	
Clear-Sky Solar Irradiance	taub	0.320	0.354	0.402	0.442	0.414	0.384	0.385	0.393	0.365	0.374	0.334	0.318	
	taud	2.315	2.235	2.151	2.083	2.241	2.358	2.368	2.331	2.405	2.339	2.411	2.382	
	Ebn at Noon	263	269	267	264	275	282	281	276	278	259	257	255	
	Edh at Noon	29	36	43	50	43	39	38	38	33	32	26	25	
All-Sky Solar Radiation	RadAvg	620	918	1166	1536	1861	2135	2157	1957	1561	1044	720	525	
	RadStd	74	125	113	183	171	179	127	117	107	91	91	66	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.17	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153	+106

Nomenclature: See separate page